# COL362/632 Introduction to Database Management Systems

## Transactions

Kaustubh Beedkar

Department of Computer Science and Engineering
Indian Institute of Technology Delhi

# Admistrivia

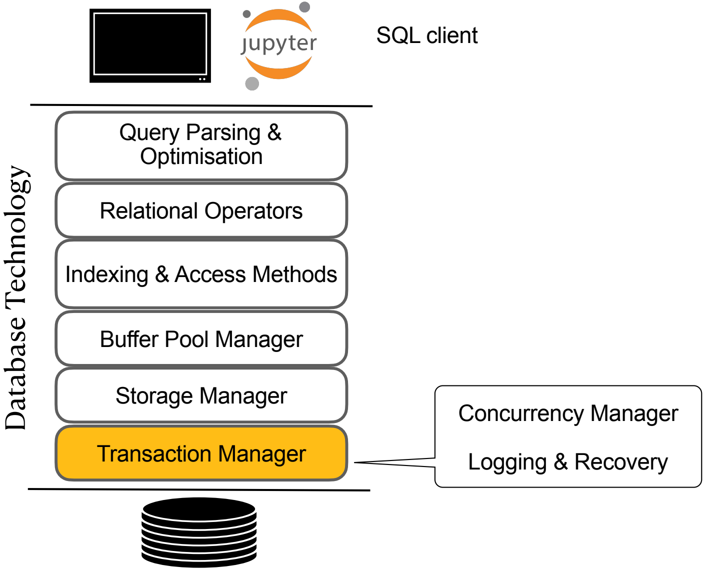
**Quiz-05**

- Monday 6th April

**Assignment**

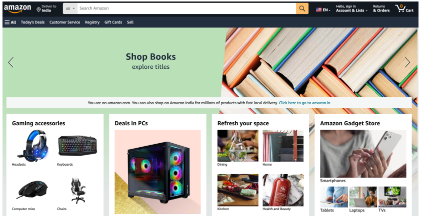
- First submission 13th – 15th April
- Subsequently (only for correct ones) submit as many number of times until 27th April
- Evaluation
  - Simple workload – for correctness
  - Workload-1 (visible)
  - Workload-2 (hidden)
- Grading
  - 30% for correctness
  - Rest relative grading (TBD after first submission)

# Course Overview



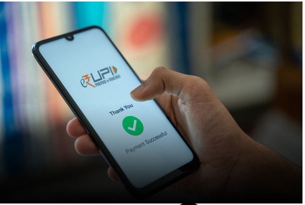
# Transactions

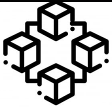
- Major component of database systems
- Critical for many applications




**IRCTC's payment gateway gains traction, handles 125,000 transactions daily**

*IRCTC introduces Autopay feature to ensure more reliable ticket booking*

## Cross shard transactions at 10 million requests per second

// By Daniel Tahara • Nov 09, 2018

Dropbox stores petabytes of metadata to support user-facing features and to power our production infrastructure. The primary system we use to store this metadata is named **Edgestore** and is described in a previous blog post, (Re)Introducing Edgestore. In simple terms, Edgestore is a service and abstraction over thousands of MySQL nodes that provides users with strongly consistent, transactional reads and writes at low latency.

WIKIPEDIA
The Free Encyclopedia

Create account Log in

# Jim Gray (computer scientist)

30 languages

Article Talk — Read Edit View history Tools

From Wikipedia, the free encyclopedia

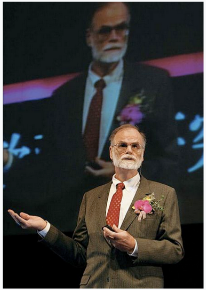
**James Nicholas Gray** (1944 – declared dead in absentia 2012) was an American computer scientist who received the Turing Award in 1998 "for seminal contributions to database and transaction processing research and technical leadership in system implementation".[4]

## Early years and personal life [ edit ]

Gray was born in San Francisco, the second child of Ann Emma Sanbrailo, a teacher, and James Able Gray, who was in the U.S. Army; the family moved to Rome, Italy, where Gray spent most of the first three years of his life; he learned to speak Italian before English. The family then moved to Virginia, spending about four years there, until Gray's parents divorced, after which he returned to San Francisco with his mother. His father, an amateur inventor, patented a design for a ribbon cartridge for typewriters that earned him a substantial royalty stream.[2]

After being turned down for the Air Force Academy he entered the University of California, Berkeley as a freshman in 1961. To help pay for college, he worked as a co-op for General Dynamics, where he learned to use a Monroe calculator. Discouraged by his chemistry grades, he left Berkeley for six months, returning after an experience in industry he later described as "dreadful".[2] Gray earned his B.S. in engineering mathematics (Math and

**Jim Gray**



Gray in 2006

| | |
|---|---|
| **Born** | James Nicholas Gray<br>January 12, 1944[1] |

# Transactions

- A transaction is a unit of work comprising sequence of updates with the **all-or-nothing** property
- Example transaction
  - read(A)
  - A := A - 50
  - write(A)
  - read(B)
  - B := B + 50
  - write(B)
- Two key challenges
  1. Failures (e.g., hardware, software, network, etc.)
  2. Concurrency (multiple transactions at the same time)

# ACID Properties

**A**tomicity

Either all or nothing: The database must ensure that updates of a partially executed transaction are not reflected in the database

**C**onsistency

A transaction as a whole must not violate integrity constraints

**I**solation

Transactions must appear to execute one after the other in sequence

**D**urabaility

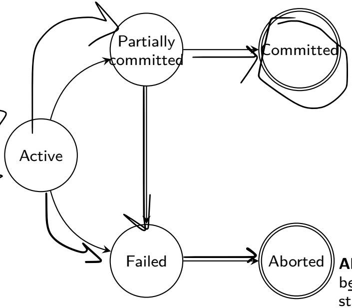
Updates to the database must persist after transaction is completed, even if there are failures

Example transaction

1. read(A)
2. A := A - 50
3. write(A)
4. read(B)
5. B := B + 50
6. write(B)

# Transaction States

**Partially committed:** state after final statement has been executed




**Committed:** After transaction has successfully completed

**Initial State:** the transaction stays in this state while it is executing

**Aborted:** After transaction has been rolled back and database state has been restored.

Either restart

or, kill the transaction

**Failed:** After normal execution can no longer proceed

# ACID Properties

**Why is hard to provide ACID properties?**

- Concurrent transactions
  - Improved throughput & resource utilization
  - Reduced waiting time (low latency)
  - But, Isolation problems!
- Failures
  - Can happen at anytime
  - Atomicity and Durability problems!

# Schedules

A **schedule** is a sequence of instructions that specify the chronological order in which instructions are executed.

# Serial Schedule

**Examples**

A serial schedule: $T_1$ is followed by $T_2$

| $T_1$ | $T_2$ |
|---|---|
| read($A$) | |
| $A := A - 50$ | |
| write($A$) | |
| read($B$) | |
| $B := B + 50$ | |
| write($B$) | |
| commit | |
| | read($A$) |
| | $temp := A * 0.1$ |
| | $A := A - temp$ |
| | write($A$) |
| | read($B$) |
| | $B := B + temp$ |
| | write($B$) |
| | commit |

A serial schedule: $T_2$ is followed by $T_1$

| $T_1$ | $T_2$ |
|---|---|
| | read($A$) |
| | $temp := A * 0.1$ |
| | $A := A - temp$ |
| | write($A$) |
| | read($B$) |
| | $B := B + temp$ |
| | write($B$) |
| | commit |
| read($A$) | |
| $A := A - 50$ | |
| write($A$) | |
| read($B$) | |
| $B := B + 50$ | |
| write($B$) | |
| commit | |

# Concurrent Schedule

A **serializable** schedule

| $T_1$ | $T_2$ |
|---|---|
| read($A$) | |
| $A := A - 50$ | |
| write($A$) | |
| | read($A$) |
| | $temp := A * 0.1$ |
| | $A := A - temp$ |
| | write($A$) |
| read($B$) | |
| $B := B + 50$ | |
| write($B$) | |
| commit | |
| | read($B$) |
| | $B := B + temp$ |
| | write($B$) |
| | commit |

A **non-serializable** schedule

| $T_1$ | $T_2$ |
|---|---|
| read($A$) | |
| $A := A - 50$ | |
| | read($A$) |
| | $temp := A * 0.1$ |
| | $A := A - temp$ |
| | write($A$) |
| | read($B$) |
| write($A$) | |
| read($B$) | |
| $B := B + 50$ | |
| write($B$) | |
| commit | |
| | $B := B + temp$ |
| | write($B$) |
| | commit |

# Serializable Schedules

- Database must ensure that the schedule is serializable
- Different notions
  - Conflict serializability
  - View serializability
- Simplified view
  - Never commute individual instructions
  - Only care about reads and writes

# Conflict Serializability

**Conflicts**

Two instructions conflict iff both access the same data item and at least one of them wrote the data item

- Read-Write
- Write-Read
- Write-Write

Conflicts enforces a temporal order!

**Definition (Conflict Serializable)**

A schedule is conflict serializable if it can be transformed into a serial schedule by a series of swappings of adjacent non-conflicting instructions

# Conflict Serializability (Example)

| $T_1$ | $T_2$ |
|---|---|
| read($A$) | |
| write($A$) | |
| | read($A$) |
| | write($A$) |
| read($B$) | |
| write($B$) | |
| | read($B$) |
| | write($B$) |

| $T_1$ | $T_2$ |
|---|---|
| read($A$) | |
| write($A$) | |
| | read($A$) |
| read($B$) | |
| | write($A$) |
| write($B$) | |
| | read($B$) |
| | write($B$) |

| $T_1$ | $T_2$ |
|---|---|
| read($A$) | |
| write($A$) | |
| read($B$) | |
| write($B$) | |
| | read($A$) |
| | write($A$) |
| | read($B$) |
| | write($B$) |

# Conflict Serializability

**Testing for conflict serializability**

- Precedence graph
  - A graph with a node for each transaction $T_i$, and
  - an edge $T_i \rightarrow T_j$ whenever an instruction $I_i$ in $T_i$ conflicts with, and comes before an instruction $I_j$ in $T_j$

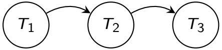
- A schedule is serializable iff the precedence graph is acyclic

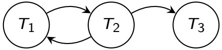
# Conflict Serializability

**Testing for conflict serializability** Q: which of the following schedules is conflict serializable?

| $T_1$ | $T_2$ | $T_3$ |
|---|---|---|
| | read(A) | |
| read(B) | | |
| | write(A) | |
| | | read(A) |
| write(B) | | |
| | | write(A) |
| | read(B) | |
| | write(B) | |

| $T_1$ | $T_2$ | $T_3$ |
|---|---|---|
| | read(A) | |
| read(B) | | |
| | write(A) | |
| | read(B) | |
| | | read(A) |
| write(B) | | |
| | | write(A) |
| | write(B) | |



# Dealing with Failures

**What could go wrong here?**

| $T_1$ | $T_2$ |
|---|---|
| read(A) | |
| write(A) | |
| | read(A) |
| | commit |
| read(B) | |
| abort | |

- Cannot abort $T_1$ because we cannot undo $T_2$
- This schedule is **not recoverable**

# Recoverable Schedules

- A schedule is **recoverable** if
  - it is conflict serializable, and
  - $T_j$ reads data item written by $T_i$, then commit of $T_i$ must appear before commit of $T_j$

| $T_1$ | $T_2$ |
|---|---|
| read(A) | |
| write(A) | |
| | read(A) |
| read(B) | |
| commit | |
| | commit |

- If $T_1$ aborts, then $T_2$ also aborts

# Cascading Aborts

- If a transaction $T$ aborts, abort all transactions $T'$ that have read data items written by $T$
- Can lead to the undoing of significant amount of work

| $T_1$ | $T_2$ | $T_3$ |
|---|---|---|
| read(A) | | |
| read(B) | | |
| write(A) | | |
| | read(A) | |
| | write(A) | |
| | | read(A) |
| abort | | |

- If $T_1$ fails, then $T_2$ and $T_3$ must also be rolled back
- It is desirable to have schedules that are **cascadeless**
  - Whenever a transaction $T$ reads a data item written by transaction $T'$, then commit of $T'$ must happen before read of $T$

# Recap of Schedules

**Serializability**

- Serial
- Serializable
- Conflict or view serializable

**Recoverability**

- Recoverable
- Cascadeless

# Recap of Schedules